

Maverick Luke

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EDUCATION

University of Toronto

Toronto, Ontario

HBSc, Specialist in Data Science and Major in Computer Science

June 2026

- CGPA: 3.79, **Dean's List** 2024, 2025 and 2026

PROJECTS

BeatManager ([github](#))

June 2026

Unreal Engine 5, C++

Personal Project

- Built a rhythm game in **Unreal Engine 5** and **C++**, using the **Quartz audio-render clock** (PlayQuantized, beat-fraction timestamps) to keep beat timing sample-accurate and drift-free instead of frame-based **DeltaTime**
- Diagnosed and fixed startup jitter and a quantization bug using custom **DSP-time instrumentation**, in a decoupled **GameMode / PlayerController / HUD / Actor** architecture with delegate-based beat events

Ghouls, Grooves, and Garlic ([itch.io](#))

Sep 2025 - Present

Unity, C#

Toronto, Ontario

- **2nd Best Game Overall** at **LevelUp 2026**, a rhythm-action game built in **Unity (C#)** with a 10-person cross-functional team for the CSC404 game design capstone
- Architected core **game manager** and centralized state control coordinating scene flow across the full game loop, plus an **extensible boss state machine** with a modular attack-pattern system that accelerated gameplay iteration
- Integrated **FMOD** into the Unity project, reimplementing the core audio system and beat-synchronization handling to route music and gameplay timing through FMOD's event system, with seamless cross-fading across tutorial-to-gameplay transitions
- Owned end-to-end implementation of the **tutorial level** (event scripting, dialogue triggers, music transitions) with a designer-authored **ScriptableObject** dialogue system

Ant Colony Simulation ([github](#))

Jan 2026

C++, Ubisoft NEXT API

Ubisoft NEXT

- Solo submission to **Ubisoft NEXT 2025-2026** simulating two rival ant colonies with pheromone pathfinding and emergent swarm behaviors at **2000 entities** in real-time, built end-to-end in **C++** on the framework's primitive draw calls
- Designed cache-friendly **Entity Component System (ECS)** with **Structure of Arrays (SoA)** memory layout and **O(1) component lookups** via per-entity bitmasks, plus a **uniform spatial grid** with a templated lambda interface for **zero-allocation** neighbor queries
- Designed **5-state FSM** for ant AI (WANDER, FOLLOW_TRAIL, FORAGE, ATTACK, FLEE) and a **3-layer pheromone system** (HOME, FOOD, ALARM) producing emergent foraging, combat coordination, and team-specific homing
- Optimized the AI hot loop with **sin/cos lookup tables**, **staggered pheromone decay**, **frustum culling**, and **screen-space LOD** selecting render complexity by entity size and zoom

EXPERIENCE

Full-Stack Developer → Tech Lead

Aug 2025 - Present

Ontario Association of Physics Teachers (OAPT), University of Toronto

Toronto, Ontario

- Promoted to **Tech Lead** after 5 months, leading a team of **7 developers** and owning end-to-end architecture of a contest platform designed for **660,000+ students** across **883+ Ontario schools**
- Built **backend services** with **Node.js/Express.js**, **PostgreSQL**, and **Prisma ORM**, including a **RESTful API** (20+ endpoints), secure authentication (**JWT**, **bcrypt**), **RBAC**, and player/user data management

SKILLS

Programming Languages: C++, C#, JavaScript/TypeScript, Python, Java, C, MIPS Assembly

Game Engines & Frameworks: Unity, Unreal Engine, Ubisoft NEXT

Game Programming: ECS, Gameplay Systems, AI/FSM, Spatial Partitioning, Game Loop Architecture, Component-Based Design, Data-Driven Design, ScriptableObjects, Coroutines

Audio & Graphics: FMOD, Quartz, 2D Camera Systems, Frustum Culling, Level of Detail (LOD), Vector Math

Performance & Optimization: Cache-Friendly Memory Layout, Structure of Arrays (SoA), Lookup Tables, Zero-Allocation Patterns

Web & Backend: Next.js, Node.js, Express.js, PostgreSQL, Prisma ORM, RESTful APIs, JWT, RBAC, Docker, AWS

Tools & Workflow: Git/GitHub, Visual Studio, Unity Editor, Trello, Slack, Linux/Unix, Shell Scripting

AWARDS & ACTIVITIES

Awards: Indonesia Maju Scholarship

Teaching: Teaching Assistant for CSC309 (Programming on the Web), University of Toronto

Leadership: Led Recognized Study Group for CSC165 (Mathematical Expression and Reasoning for CS)

Languages: Indonesian (native), English (fluent)